Weighted tropical Chow reconstruction for space curves

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Abstract

We prove that the weighted tropical Chow transform determines a tropical curve in $\mathbb{R}^4/\mathbb{R}\mathbf{1}$, without requiring its support to be a fan. More generally, the transform is injective on finite rational polyhedral one-chains with signed rational weights, whether or not they are balanced. The proof separates contributions according to their affine three-dimensional spans. For edges outside the coordinate directions, small parameters in a suitable incidence cone localize every competing source point. For coordinate directions, the remaining contributions split into a term independent of a thickness parameter and a thin band. The thin band tends to zero locally in L^1, while its normalized measure converges to the source length measure. This eliminates the remaining kernel without any general-position or realizability assumption.

1 Introduction

The tropical Chow construction associates to a tropical variety a weighted incidence cycle in a tropical Grassmannian. For a curve $\Sigma \subset \mathbb{R}^4/\mathbb{R}\mathbf{1}$, this cycle parametrizes tropical lines meeting Σ, with multiplicities determined by a weighted Minkowski sum. In the public arXiv version 2 of his paper, Tripoli formulated the injectivity problem and proved the space-curve case when the support is a fan [3, Conjecture 3.4 and Theorem 3.9]. All numbered references to that paper below use this same public version.

Our purpose is to remove the fan hypothesis for curves with finitely many polyhedral cells. All equalities below retain weights and are taken after common subdivision. In particular, positions in the ambient space are part of the data.

Theorem 1.1. *Let Σ and Σ' be finite rational polyhedral tropical curves in $\mathbb{R}^4/\mathbb{R}\mathbf{1}$, with positive integral weights. Then*

$$Z_\Sigma = Z_{\Sigma'} \quad \Longrightarrow \quad \Sigma = \Sigma'.$$

The curves may have bounded edges, distinct vertices, and more than one connected component.

Finiteness means a finite polyhedral structure, not compact support. Thus rays and complete affine lines are allowed. Rational polyhedra are taken with respect to the indicated quotient lattices. We work in the finite-coordinate tropical tori, without additional strata at infinity.

The proof uses a linear extension of the Chow transform to signed polyhedral chains. Such chains are unoriented weighted polyhedra modulo subdivision; balancing is not imposed. Write $\mathcal{C}_d(V)$ for their $\mathbb{Q}$-vector space in a real vector space V with a fixed lattice. Section 2 defines a map

$$T : \mathcal{C}_1(N_\mathbb{R}) \longrightarrow \mathcal{C}_3(M_\mathbb{R}), \qquad N = \mathbb{Z}^4/\mathbb{Z}\mathbf{1}, \quad M = \mathbb{Z}^6/\mathbb{Z}\mathbf{1}.$$

Theorem 1.2. *The map T is injective.*

This stronger statement is useful because separating contributions in the target does not preserve balancing. We therefore carry out that separation in the vector space of chains, and recover Theorem 1.1 by restricting to balanced inputs.

In the proof of Lemma 3.8, Tripoli already extends the transform to rationally weighted, possibly unbalanced fans supported on the eight rays $\mathbb{R}_{\geq 0}e_i$ and $\mathbb{R}_{\geq 0}(-e_i)$, $i = 0, \ldots, 3$, all based at the origin, and proves its injectivity [3]. Our extension allows arbitrary finite affine one-chains; thin-band recovery handles coordinate edges in arbitrary positions.

Although every tropical cycle in a vector space can be expressed as a sum of translated fan cycles [1, Theorem 5.4], this does not by itself reduce the question to injectivity on fans. Images of distinct translated summands may cancel. Our argument controls precisely these overlaps; it does not require choosing a decomposition into fan cycles.

There are two parts to the proof. First, different affine three-dimensional spans in the target can meet only in lower dimension. Within a suitable span, the contributions of an edge not parallel to a coordinate direction come from just one incidence cone. A short-parameter argument then rules out cancellation by distant source points. Second, for coordinate directions, several cones do contribute to the same span. Their weight function has the form

$$K_t = H + C_t,$$

where H contains all side-cone contributions and does not depend on t. The central contribution C_t is a thin band swept out by translating the source through a segment of length t. A boundary limit first removes H and then recovers the source measure.

Weights are essential: the injectivity statement for incidence supports alone already fails for fan curves [3, Example 3.5]. The result concerns the full weighted Chow cycle, rather than a Chow polytope or its normal complex; see [2] for the latter construction. Our argument establishes uniqueness. It does not assert an algorithm, or a complexity bound, for reconstructing an arbitrary input from its output.

2 The chain transform and the incidence fan

2.1 Polyhedral chains and weights

For a rational affine polyhedron P in a lattice vector space V with lattice L, put

$$\operatorname{lin}(P) = \operatorname{span}_{\mathbb{R}}(P - P), \qquad L_P = L \cap \operatorname{lin}(P).$$

In particular, the lattice of an affine cell is its direction lattice, not the group generated by its affine integral points.

Definition 2.1. The vector space $\mathcal{C}_d(V)$ consists of finite rational linear combinations of symbols $[P]$ for rational d-dimensional polyhedra, modulo subdivision: if P is subdivided into d-dimensional cells P_i, then $[P] = \sum_i [P_i]$. Cells of dimension less than d are discarded. Equivalently, an element is a finite polyhedral complex with signed rational weights on its d-cells, modulo common subdivision and removal of zero-weight cells. No orientation or balancing condition is included.

For a chosen representative of a one-chain D, write $D^{(1)}$ for its one-dimensional cells. Such a chain is balanced if, at every vertex x,

$$\sum_{e \ni x} w_e u_{e,x} = 0,$$

where $u_{e,x}$ is the primitive direction of e pointing away from x. One may subdivide complete lines to state this condition. A tropical curve in this paper is an effective balanced one-chain with integral weights; its vertices carry no additional zero-dimensional weights. The support is the closure of the nonzero-weight one-cell interiors. At a point in the relative interior of a one-cell, a sufficiently small neighborhood of this support consists of that cell alone.

Let e_i denote the images of the standard basis of $\mathbb{Z}^4$ in N. Use the order $01, 02, 03, 12, 13, 23$ for the coordinates of M, and write E_{ij} for its standard generators. Pair subscripts are unordered, so $E_{ij} = E_{ji}$ and the same convention applies to pair coordinates. Define

$$\varphi : N \longrightarrow M, \qquad [u_0, u_1, u_2, u_3] \longmapsto [u_i + u_j]_{i<j}, \qquad f_i = \varphi(e_i).$$

The map is well defined. It is injective and has saturated image: for distinct i, j and a third index k,

$$u_i - u_j = \varphi(u)_{ik} - \varphi(u)_{jk}.$$

Consequently, if $\varphi(u)$ represents an integral point of M, all differences $u_i - u_j$ are integral, so u represents a point of N.

Let Γ_0 be the tropical incidence fan of lines through the origin, with its usual multiplicities. We use the minimum convention for tropicalization; in this convention the standard tropical line has rays in directions e_0, e_1, e_2, e_3. The constant-coefficient incidence space may be taken over $\mathbb{C}$, and rational valuations realized over the Puiseux series field. Its explicit incidence cones are given below. For a one-cell e and a two-cone τ of Γ_0, set

$$P_{e,\tau} = \varphi(e) + \tau.$$

If this polyhedron has dimension three, define

$$m(e, \tau) = \big[M \cap \operatorname{lin}(P_{e,\tau}) : \varphi(N_e) + M_\tau\big]. \tag{1}$$

This is a positive finite integer. For a chain $D = \sum_e w_e[e]$, define

$$T(D) = \sum_{\substack{e \in D^{(1)},\ \tau \in \Gamma_0^{(2)} \\ \dim P_{e,\tau} = 3}} w_e w_\tau m(e, \tau)[P_{e,\tau}], \qquad w_\tau = 1. \tag{2}$$

Overlaps in this sum are added after common subdivision.

Lemma 2.2. *Formula* (2) *defines a linear map on* $\mathcal{C}_1(N_\mathbb{R})$ *and is unchanged by subdivision of the incidence fan. For a balanced input it agrees with the weighted Minkowski transform* $Z_D = \varphi(D) + \Gamma_0$.

Proof. On a product $e \times \tau$ with three-dimensional image, the addition map $(x, g) \mapsto \varphi(x) + g$ is injective on the direction space: both domain and image have dimension three. A subdivision of the product therefore subdivides its image, with the same direction lattices and the same index (1). Images of subdivision boundaries have dimension at most two. If the original image has smaller dimension, every subdivided image does too, and all are discarded. This proves well-definedness and subdivision invariance; linearity follows from the weight formula. On balanced chains the formula is the standard full-rank weighted Minkowski sum, as in [3, Section 2, equation (5) and Definition 2.7]. □

Remark 2.3. For a nonbalanced chain we use the explicit linear map (2), not a claim that the input is a tropical cycle. There is no additional division by a covering degree in this definition.

2.2 The fifteen cones

For a partition $\{a,b\} \sqcup \{c,d\} = \{0,1,2,3\}$, write

$$h_{ab} = e_a + e_b, \qquad H_{ab} = \operatorname{span}_{\mathbb{R}}(e_a, e_b).$$

An identity used throughout is

$$\varphi(h_{ab}) = f_a + f_b = E_{ab} - E_{cd}. \tag{3}$$

Lemma 2.4. *The fan Γ_0 admits a structure with the following fifteen maximal cones, each of dimension two and weight one:*

$$\sigma_{ab|cd} = \operatorname{pos}(E_{ab}, E_{cd}) \qquad \text{for the three unordered partitions } ab|cd, \tag{4}$$

$$\tau_{ij;k} = \operatorname{pos}(E_{ij}, -f_k) \qquad \text{for } k \notin \{i,j\}. \tag{5}$$

In each case the displayed generators form a basis of the saturated direction lattice of the cone.

Proof. The incidence linear space has Plücker coordinates $p_{ij} = z_j - z_i$: a line through $[1:1:1:1]$ is spanned by that point and $[z_0 : z_1 : z_2 : z_3]$. Set $w_{ij} = \operatorname{val}(z_j - z_i)$ and normalize $\min_{i<j} w_{ij} = 0$. For any three labels, the minimum of their three pair valuations is attained at least twice. Thus for each $r \geq 0$, the relation $i \sim_r j$ defined by $w_{ij} > r$, with $i \sim_r i$, is an equivalence relation. Its nontrivial classes form nested proper clusters as r increases.

On four labels there are at most two nontrivial proper clusters in a maximal such hierarchy. They are either two disjoint two-element clusters, or a three-element cluster with a two-element subcluster. Indeed, two proper clusters which intersect are nested, a proper cluster has at most three labels, and a three-element cluster contains at most one two-element cluster in a nested hierarchy. The first pattern gives (4). For the second, the valuation vector of the three-element cluster complementary to k is

$$\sum_{i<j,\ i,j\neq k} E_{ij} = -f_k,$$

and its subcluster contributes E_{ij}, giving (5). Hierarchies with fewer clusters give faces. Conversely, the interiors of the two representative cones are realized, for rational $\alpha, \beta, r, s > 0$, by

$$(z_0, z_1, z_2, z_3) = (0, t^\alpha, 1, 1 + t^\beta), \qquad (z_0, z_1, z_2, z_3) = (0, t^{r+s}, 1, t^s),$$

respectively. These valuation vectors are $\alpha E_{01} + \beta E_{23}$ and $rE_{01} - sf_2$ in $M_{\mathbb{R}}$. Permuting the labels and taking closures gives all fifteen cones.

For completeness, their multiplicities can be read from two initial ideals. The incidence ideal is generated by the three independent linear relations

$$p_{12} - p_{02} + p_{01}, \qquad p_{13} - p_{03} + p_{01}, \qquad p_{23} - p_{03} + p_{02}.$$

At an interior point of $\sigma_{01|23}$ its initial ideal is

$$(p_{12} - p_{02},\ p_{13} - p_{03},\ p_{02} - p_{03}),$$

and at an interior point of $\tau_{01;2}$ it is

$$(p_{12} - p_{02},\ p_{13} - p_{03},\ p_{23} + p_{02}).$$

The displayed initial forms lie in the respective initial ideals. In each case they generate an ideal of three independent linear forms, with the same ordinary graded Hilbert series as the incidence ideal. Preservation of that Hilbert series under taking weight initial ideals shows that these are the entire initial ideals. Both define reduced irreducible linear spaces with nonempty coordinate-torus intersections, so their torus degenerations have multiplicity one. Permutations of labels give the same conclusion for every maximal cone. The fan description, including these weights, is also recorded in [3, Section 3.1, Figure 3].

For the lattice assertion it suffices, by permutations of the four labels, to check (E_{01}, E_{23}) and $(E_{01}, -f_2)$. Identify M with $\mathbb{Z}^5$ by subtracting the 23 coordinate and then deleting it. The first pair has a 2×2 minor of absolute value one in rows $01, 02$, and the second in rows $01, 03$. Thus each pair generates its saturated rank-two lattice. □

To keep track of the three partitions, consider the quotient map

$$\pi : M_{\mathbb{R}} \longrightarrow \mathbb{R}^2, \qquad p \longmapsto (p_{01} + p_{23} - p_{02} - p_{13},\, p_{01} + p_{23} - p_{03} - p_{12}). \tag{6}$$

Its kernel is $\varphi(N_{\mathbb{R}})$: the inclusion follows from the formula for φ, and both spaces have dimension three. The images of E_{01}, E_{02}, E_{03} are respectively $(1,1), (-1,0), (0,-1)$, while complementary pairs have equal images. In particular the three partition directions are pairwise noncollinear.

The partition criterion in the following lemma is equivalent to [3, Lemma 3.7]. Its choice of central cone and separation of linear spans follow the proof of [3, Theorem 3.9]; we give a self-contained argument in our notation. The passage to arbitrary affine sources will additionally require grouping by exact affine span (Lemma 3.1) and localizing competing source points through the equality $\alpha' + \beta' = \alpha + \beta$ (Proposition 3.2, equation (9)).

Lemma 2.5. *Let $v \in N_{\mathbb{R}}$ be nonzero and not parallel to any e_i. There is a partition $ab|cd$ such that $v \notin H_{ab} \cup H_{cd}$. For this partition the space*

$$U = \operatorname{span}_{\mathbb{R}}(\varphi(v), E_{ab}, E_{cd})$$

has dimension three, contains none of the f_i, and contains exactly one maximal cone of Γ_0, namely $\sigma_{ab|cd}$. Moreover,

$$U \cap \varphi(N_{\mathbb{R}}) = \varphi(\operatorname{span}_{\mathbb{R}}(v, h_{ab})). \tag{7}$$

Proof. Membership in H_{ab} means that the complementary two coordinates of a representative of v are equal. Since v is not a coordinate direction, no three of its four coordinates are equal. We can therefore partition the four indices into two pairs within each of which the coordinates differ. This gives the required partition.

The vectors v, h_{ab} are independent, since h_{ab} belongs to both H_{ab} and H_{cd}. Equations (3) and (6) give the dimension statement and (7). If $f_i \in U$, then $e_i \in \operatorname{span}_{\mathbb{R}}(v, h_{ab})$. As e_i is independent of h_{ab}, this would imply $v \in H_{ab}$ for $i \in \{a, b\}$, or $v \in H_{cd}$ otherwise. Hence no side cone is contained in U. The quotient directions in (6) exclude the other central cones. □

Lemma 2.6. *For a pair a, b with complementary pair c, d, put*

$$U_{ab} = \operatorname{span}_{\mathbb{R}}(f_a, f_b, E_{cd}).$$

Then $U_{ab} \cap \varphi(N_{\mathbb{R}}) = \varphi(H_{ab})$, and $f_i \in U_{ab}$ if and only if $i \in \{a, b\}$. The maximal cones of Γ_0 contained in U_{ab} are exactly

$$\sigma_{ab|cd}, \quad \operatorname{pos}(E_{cd}, -f_a), \quad \operatorname{pos}(E_{cd}, -f_b). \tag{8}$$

The vectors f_a, f_b, E_{cd} form a basis of $M \cap U_{ab}$. Each full-rank pairing of a source direction e_a or e_b with a cone in (8) has index one in (1).

Proof. The intersection statement follows by applying π, and the assertion about the f_i follows since any three of the e_i form a basis of N. The quotient directions restrict possible cones to the partition $ab|cd$. Among its side cones, exactly those displayed have their coordinate generator in U_{ab}.

By symmetry take $ab = 01$, $cd = 23$. In the quotient coordinates used in Lemma 2.4, the columns f_0, f_1, E_{23} have the minor

$$\det \begin{pmatrix} 1 & 1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} = 1$$

in rows $01, 02, 12$. Thus they form a basis of the saturated lattice. In the basis f_a, f_b, E_{cd}, identity (3) reads $E_{ab} = (1, 1, 1)$. Each of the four noncollapsed source–cone pairings in (8) consequently has determinant ± 1. Lemma 2.4 identifies the cone lattices with those generated by the displayed rays, so these determinants are exactly the required indices. The other two pairings collapse to dimension two. □

3 Separating contributions and noncoordinate edges

Lemma 3.1 (Affine-span separation)**.** *Let $B = \sum_{j=1}^{r} c_j[P_j] \in \mathcal{C}_d(V)$, and fix an affine d-plane P. The function*

$$b_P = \sum_{j:\, \mathrm{aff}(P_j)=P} c_j \mathbf{1}_{P_j}$$

is determined almost everywhere by the chain B, independently of its representation. Moreover, $B = 0$ if and only if $b_P = 0$ almost everywhere for every affine span occurring in a representation. Here measure on P is Lebesgue measure on its direction space.

Proof. Every distinct affine d-plane $\mathrm{aff}(P_j)$ meets P in dimension at most $d-1$, if it meets it at all. Remove these finitely many intersections and the boundaries of the polyhedra with affine span P. On what remains, the displayed sum is exactly the coefficient of B after common subdivision. All removed sets have measure zero. Thus a zero chain gives a zero function, and subtracting two representations proves independence of the representation. Conversely, each full-dimensional cell of a common subdivision has one of the finitely many affine spans. Its coefficient is the almost-everywhere value of the corresponding function. If all these functions vanish, all cell coefficients vanish. □

Proposition 3.2. *If $T(D) = 0$, every nonzero edge of D is parallel to some e_i.*

Proof. Suppose that e has nonzero weight w_e and primitive direction v not parallel to any e_i. Choose the partition and the space U from Lemma 2.5, and write $h = h_{ab}$. Take an interior point x_0 of e and a small neighborhood in which D is just a constant-weight open segment of e. This is possible after common subdivision, since the chain has only finitely many cells. Let x vary over a smaller open segment about x_0, and set

$$z = \varphi(x) + \alpha E_{ab} + \beta E_{cd}, \qquad \alpha, \beta > 0, \quad \alpha + \beta < \varepsilon.$$

For $\varepsilon > 0$ small enough, these points form a nonempty open subset of the affine three-plane $P = \varphi(x_0) + U$.

Any summand of (2) with affine span P must use a cone contained in U. By Lemma 2.5 this cone is $\sigma_{ab|cd}$. Consider a competing representation

$$z = \varphi(y) + \alpha' E_{ab} + \beta' E_{cd}, \qquad \alpha', \beta' \geq 0,$$

with y on the support of D. Applying π and using (3) gives

$$\alpha' + \beta' = \alpha + \beta, \qquad y - x = (\alpha - \alpha')h, \qquad |\alpha - \alpha'| \le \alpha + \beta. \tag{9}$$

By reducing ε uniformly on the smaller segment, we force y into the neighborhood where the source consists only of e. Since v and h are independent, the second identity then implies $y = x$, $\alpha' = \alpha$, and $\beta' = \beta$.

Thus the sum of contributions with affine span P has, on this open set, the single coefficient $w_e m(e, \sigma_{ab|cd}) \neq 0$. All other affine spans can be ignored by Lemma 3.1. This contradicts $T(D) = 0$. □

4 Thin bands and coordinate edges

4.1 A measure lemma

Consider a finite signed rational one-chain B in $\mathbb{R}^2$ whose edges are horizontal or vertical. Its lattice-length measure is the locally finite signed Radon measure, understood locally,

$$\mu_B = \sum_{(I,\eta)} w_{I,\eta}\, \mathbf{1}_I(X)\, dX\, \delta_\eta(dY) + \sum_{(\xi,J)} w_{\xi,J}\, \delta_\xi(dX)\, \mathbf{1}_J(Y)\, dY. \tag{10}$$

The intervals may be bounded or unbounded. Endpoints have measure zero. For $s \in \mathbb{R}$, let $\tau_s(X, Y) = (X + s, Y + s)$, and for $t > 0$ put

$$\begin{aligned} C_t(X,Y) = &\sum_{(I,\eta)} w_{I,\eta} \mathbf{1}_{\{0<Y-\eta<t\}} \mathbf{1}_I(X - Y + \eta) \\ &+ \sum_{(\xi,J)} w_{\xi,J} \mathbf{1}_{\{0<X-\xi<t\}} \mathbf{1}_J(Y - X + \xi). \end{aligned} \tag{11}$$

Lemma 4.1 (Thin-band recovery)**.** *As locally finite signed Radon measures,*

$$C_t\, dX\, dY = \int_0^t (\tau_s)_* \mu_B\, ds. \tag{12}$$

As $t \downarrow 0$,

$$C_t \longrightarrow 0 \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^2), \qquad t^{-1} C_t\, dX\, dY \longrightarrow \mu_B \tag{13}$$

against all compactly supported continuous test functions. The measure integral in (12) *is understood by testing against compactly supported functions. In particular, if $H \in L^1_{\mathrm{loc}}(\mathbb{R}^2)$ is independent of t and $H + C_t = 0$ for almost every (X, Y, t) with $t > 0$, then $B = 0$.*

Proof. For a horizontal edge, the map $(x, s) \mapsto (x + s, \eta + s)$ has Jacobian of absolute value one; for a vertical edge the same holds for $(y, s) \mapsto (\xi + s, y + s)$. Change of variables gives (12), including the coefficients in (11).

If $K \subset \mathbb{R}^2$ is compact and $0 < t \le 1$, then

$$\int_K |C_t|\, dX\, dY \le \int_0^t |\mu_B|(\tau_{-s}K)\, ds \le t\, |\mu_B|\big(K + [-1, 0](1, 1)\big).$$

The last quantity is finite and tends to zero. This proves the first limit even when the source has unbounded edges. For a compactly supported continuous function g,

$$\frac{1}{t} \int g C_t\, dX\, dY = \int \left(\frac{1}{t} \int_0^t g(X + s, Y + s)\, ds \right) d\mu_B(X, Y).$$

Uniform continuity on a fixed enlarged compact set shows that this converges to $\int g\,d\mu_B$, proving the second limit.

For the last assertion, apply Fubini's theorem on a countable compact exhaustion of $\mathbb{R}^2$. It gives a single full-measure set $G \subset (0,1)$ such that $H + C_t = 0$ almost everywhere on all of $\mathbb{R}^2$ for each $t \in G$. Choose $t_n \in G$ decreasing to zero. The first limit gives $H = 0$ locally in L^1. Then $C_{t_n} = 0$, and the second limit gives $\mu_B = 0$. After common subdivision, an interior subsegment of any nonzero edge can be isolated from all other edges. A nonnegative continuous test function supported in this neighborhood and positive on the subsegment has a nonzero integral against μ_B, contradicting $\mu_B = 0$. Hence $B = 0$. □

4.2 The coordinate chart

Proposition 4.2. *If every edge of D is parallel to some e_i and $T(D) = 0$, then $D = 0$.*

Proof. Fix a pair a, b with complement c, d, and an affine plane $A = o + H_{ab}$ containing at least one source edge parallel to e_a or e_b. Set $P = \varphi(o) + U_{ab}$. By Lemma 2.6, the contributions with affine span P come exactly from the e_a- and e_b-parallel edges contained in A, paired with the three cones in (8), and with collapsed pairings omitted. Indeed, any contributing direction f_i and cone must lie in U_{ab}; its source offset then lies in A because $U_{ab} \cap \varphi(N_{\mathbb{R}}) = \varphi(H_{ab})$.

Write points of A as $o + Xe_a + Ye_b$, and let B be the horizontal and vertical chain given by these source edges. These coordinates normalize lattice length, since e_a, e_b form a basis of $N \cap H_{ab}$. Use coordinates on P given by

$$z = \varphi(o) + Xf_a + Yf_b + tE_{cd}, \qquad t > 0. \tag{14}$$

All noncollapsed indices are one by Lemma 2.6.

In the central cone, identity (3) gives

$$rE_{ab} + sE_{cd} = r(f_a + f_b) + (r + s)E_{cd}.$$

Thus, at thickness $t = r + s$, its planar displacement is (r, r) with $0 < r < t$. The total central contribution is precisely C_t from (11).

The two side cones give displacements $-rf_a$ and $-rf_b$, while their E_{cd} coefficient is t. A horizontal source edge pairs nontrivially only with the $-f_b$ side, and a vertical source edge only with the $-f_a$ side. Their total contribution is

$$H(X,Y) = \sum_{(I,\eta)} w_{I,\eta} \mathbf{1}_I(X) \mathbf{1}_{\{Y<\eta\}} + \sum_{(\xi,J)} w_{\xi,J} \mathbf{1}_{\{X<\xi\}} \mathbf{1}_J(Y). \tag{15}$$

This is a locally integrable function independent of t. It includes all distant side-cone contributions, not just those near the point being probed.

The weight function of the contributions with affine span P is therefore $K(X,Y,t) = H(X,Y) + C_t(X,Y)$, almost everywhere in the chart. Since $T(D) = 0$, Lemma 3.1 gives $K = 0$ almost everywhere. Lemma 4.1 implies $B = 0$.

For a fixed pair there are only finitely many planes A arising from the source edges. Repeating the argument for these planes and for pairs covering all four indices eliminates every edge of D. □

Proof of Theorems 1.2 and 1.1. If $T(D) = 0$, Proposition 3.2 reduces the support to coordinate directions, and Proposition 4.2 gives $D = 0$. This proves Theorem 1.2. Given the hypotheses of Theorem 1.1, take the signed chain $D = \Sigma - \Sigma'$. By Lemma 2.2, its transform is $Z_\Sigma - Z_{\Sigma'} = 0$, so $D = 0$. □

Remark 4.3. The thin-band argument uses only local finiteness of the length measure, change of variables, and uniform continuity of test functions. The conclusion does not require selecting an arbitrary slice of an almost-everywhere identity: the sequence of valid slices in Lemma 4.1 handles this explicitly.

5 A two-vertex example

Let $v = e_0 + e_1$, let $p \in N_{\mathbb{Q}}$, and let $L \in \mathbb{Q}_{>0}$. Consider the chain consisting of the bounded edge $[p, p + Lv]$, the two rays from p in directions $-e_0, -e_1$, and the two rays from $p + Lv$ in directions $-e_2, -e_3$, all of weight $m \in \mathbb{Z}_{>0}$. It is balanced at both vertices, since

$$v - e_0 - e_1 = 0, \qquad -v - e_2 - e_3 = 0.$$

The parameters can be recovered without constructing the entire output.

Proposition 5.1. *For this family the weighted Chow transform determines p, L, m.*

Proof. Put $d = \varphi(v)$ and $U = \operatorname{span}_{\mathbb{R}}(d, E_{02}, E_{13})$. The split $02|13$ satisfies the hypothesis of Lemma 2.5. Thus no f_i belongs to U, and the only incidence cone contained in U is $\sigma_{02|13}$. The output's three-dimensional part with direction space U consequently has support closure

$$S = \varphi(p) + [0, L]d + \operatorname{pos}(E_{02}, E_{13}).$$

This extraction is intrinsic to the weighted chain: contributions with different direction spaces meet it only in lower dimension, and the four unbounded source edges cannot contribute with direction space U.

The three vectors d, E_{02}, E_{13} are independent, so S is an affine image of $[0, L] \times \mathbb{R}^2_{\geq 0}$. Its two vertices are $\varphi(p)$ and $\varphi(p) + Ld$. The fixed positive direction d orders these vertices, recovering p and L. In the quotient coordinates obtained by deleting 23 after subtraction, the columns d, E_{02}, E_{13} have a determinant-one minor in rows $03, 02, 13$. The image weight is therefore m, recovering the last parameter. □

Remark 5.2. The finite hypothesis in the general theorem is used to separate affine spans and to isolate source interiors. The theorem does not address chains with infinitely many cells, higher-dimensional source varieties, or additional projective boundary strata. The example gives an explicit recovery procedure only for this family.

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